

Properties of Subspaces

Suppose U, W are subspaces of a vector space V .

Which of the following sets continue to be a subspace:

(i) $U + W := \{ u + v, u \in U, v \in W \}$ ← addition of V

YES, it is a subspace

Direct Sum of U , and W . (Verify)

(ii) $U \cap W = \{ u, \text{ s.t. } u \in U \text{ and } u \in W \}$

YES, it is a subspace

(iii) $U \cup W = \{ u, \text{ s.t. either } u \in U \text{ or } u \in W \}$

MAY NOT BE A SUBSPACE. (SEE PART LATER)

Proof of $U \cap W$ being a subspace:

Is $U \cap W$ nonempty? YES

Suppose $a, b \in U \cap W$, we need to show

$$a+b \in U \cap W.$$

$$a \in U, b \in U \Rightarrow a+b \in U \quad (U \text{ is a subspace})$$

$$a \in W, b \in W \Rightarrow a+b \in W \quad (W \text{ is a subspace})$$

$$\Rightarrow a+b \in U \cap W$$

Similarly show $\alpha \cdot a \in U \cap W$ whenever $a \in U \cap W$ and $\alpha \in \mathbb{F}$.

An Example of intersection of Subspaces

$$S_1 = \{x \in \mathbb{R}^n, \text{ s.t. } a_{11}x_1 + \dots + a_{1n}x_n = 0\}$$

$$S_2 = \{x \in \mathbb{R}^n, \text{ s.t. } a_{21}x_1 + \dots + a_{2n}x_n = 0\}$$

$\vdots$

$$S_K = \{x \in \mathbb{R}^n, \text{ s.t. } a_{k1}x_1 + \dots + a_{kn}x_n = 0\}$$

$S_1, \dots, S_K$ are subspaces

$\bigcap_{i=1}^K S_i = S_1 \cap S_2 \cap \dots \cap S_K$ is also a subspace.

$$\bigcap_{i=1}^k S_i = \left\{ x \in \mathbb{R}^n, \text{ s.t. } \begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = 0 \\ a_{21}x_1 + \dots + a_{2n}x_n = 0 \\ \vdots \\ a_{k1}x_1 + \dots + a_{kn}x_n = 0 \end{cases} \right\}$$

Note $0 \in \bigcap_{i=1}^k S_i$

FACT ABOUT UNION

Given subspaces U and W of the same vector space V , the set $U \cup W$ is a subspace if and only if at least one of following hold

$$(i) U \subseteq W \quad (ii) W \subseteq U$$

Proof: We first prove the "IF" part of the statement:

If $U \subseteq W$, $\Rightarrow U \cup W = W$ which is a subspace. Similarly if $W \subseteq U$, then $W \cup U = U$, which is a subspace.

We now prove the "ONLY IF" part of the statement.

Suppose $U \cup W$ is a subspace. We prove by contradiction. Suppose neither U is a subspace of W nor W is a subspace of U .

$\Rightarrow$ there exists $u \in U$ s.t. $u \notin W$
and there exists $w \in W$ s.t. $w \notin U$

Observe that $u, w \in U \cup W$

Since $U \cup W$ is a subspace,

$$\Rightarrow u + w \in U \cup W$$

$\Rightarrow u+w$ must belong to at least one of $\mathcal{U}$ or $\mathcal{W}$.

Suppose $u+w \in \mathcal{U}$

notice $u_I \in \mathcal{U}$ (since $\mathcal{U}$ is subspace)

$\Rightarrow (u+w) + u_I \in \mathcal{U}$ (since $\mathcal{U}$ is a subspace)

$\Rightarrow w + \underbrace{u+u_I}_{=0} \in \mathcal{U}$

$\Rightarrow w \in \mathcal{U} \Rightarrow$ contradicts $w \notin \mathcal{U}$

If $u+w \in \mathcal{W}$, using a similar argument, establish that $u \in \mathcal{W}$,
which again contradicts $u \notin \mathcal{W}$

$\Rightarrow$ It cannot be that neither $\mathcal{U} \subseteq \mathcal{W}$
nor $\mathcal{W} \subseteq \mathcal{U}$. (Proved)

Linear Independence

Given a vector space V (over field F), a set of non-zero vectors $v_1, \dots, v_k$ are said to be linearly independent if

$$\sum_{i=1}^k \alpha_i v_i = 0 \iff \alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_k = 0$$

IMPLICATIONS OF LINEAR INDEPENDENCE

① Suppose $v_1, v_2, \dots, v_k \in \mathbb{F}^n$ are linearly independent.

Define $V = [v_1 \ v_2 \ \dots \ v_k] \in \mathbb{F}^{n \times k}$

Then $N(V) = \{0\}$

$$\sum_{i=1}^k \alpha_i v_i = [v_1 \ \dots \ v_k] \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_k \end{bmatrix}$$

$$= V \cdot \alpha$$

$$, \alpha = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_k \end{bmatrix}$$

$$\sum_{i=1}^k \alpha_i v_i = 0 \Leftrightarrow \alpha_1 = \dots = \alpha_k = 0$$

$$\Leftrightarrow \forall \alpha = 0 \Leftrightarrow \alpha = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\Leftrightarrow \boxed{N(v) = \{0\}}$$

② UNIQUE REPRESENTATION:

Suppose $S = \text{span} \{v_1, \dots, v_k\}$ where $v_1, \dots, v_k$ are linearly independent.

Let $u \in S$. Then there exist $\alpha_1, \dots, \alpha_k \in F$ s.t.

$$u = \sum_{i=1}^k \alpha_i v_i$$

Is there another set of scalars $\beta_1, \dots, \beta_k \in F$

s.t.

$$u = \sum_{i=1}^k \beta_i v_i \quad ?$$

Suppose there is. Then

$$u = \sum_{i=1}^k \alpha_i v_i = \sum_{i=1}^k \beta_i v_i \Rightarrow \sum_{i=1}^k (\alpha_i - \beta_i) v_i = 0$$

Since $\{v_i\}_{i=1}^k$ are linearly independent, we must have $\alpha_i - \beta_i = 0 \Rightarrow \alpha_i = \beta_i, i=1, 2, \dots, k.$

Therefore, each vector in S has a unique representation as a linear combination of $v_1, \dots, v_k$.

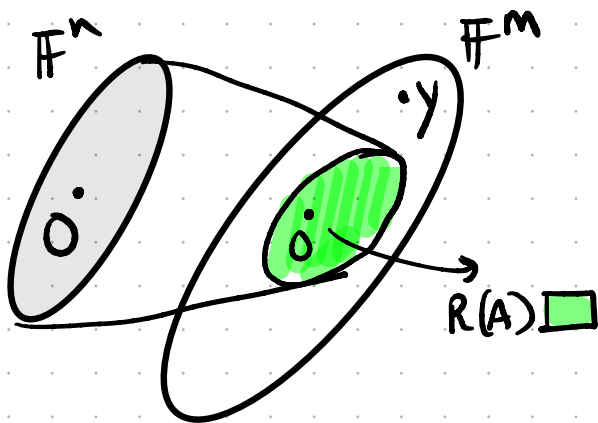
3 SOLUTIONS TO SYSTEMS OF LINEAR EQUATIONS :-

Given $y \in \mathbb{F}^m$, $A \in \mathbb{F}^{m \times n}$, solve for $x \in \mathbb{F}^n$:

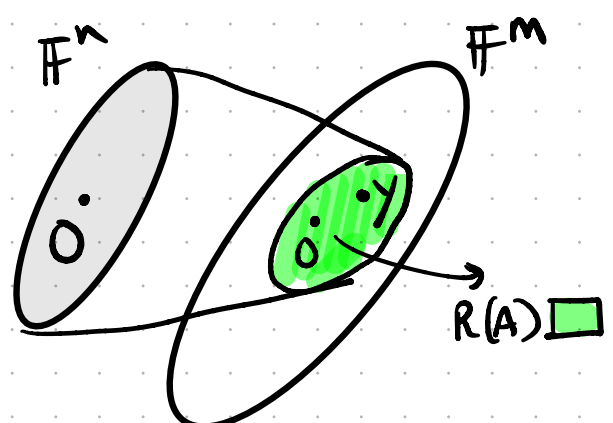
$$y = Ax$$

Does solution exist?

Is there $x \in \mathbb{F}^n$ s.t. $y = Ax$?
YES, if and only if $y \in \mathcal{R}(A)$



No Solution



Solution Exists

Is solution unique?

Depends on $N(A)$

Suppose $y \in R(A)$ (hence solution exists).

and $N(A) = \{0\}$

Then $y = Ax$ has unique solution.

Why?

If not, suppose x_1, x_2 be two solutions to

$$y = Ax \Rightarrow Ax_1 = Ax_2 \Rightarrow A(x_1 - x_2) = 0$$

$$\Rightarrow x_1 - x_2 \in N(A).$$

$$\text{But } N(A) = \{0\} \Rightarrow \boxed{x_1 = x_2}$$

Similarly, it can be shown that if $y = Ax$ has unique solution, then $N(A) = \{0\}$.

If $N(A) \neq \{0\}$, there exists $v \neq 0$

$$\text{s.t. } Av = 0$$

If x_0 is a solution to $y = Ax$, then

$x_0 + v$ ($\neq x_0$, since $v \neq 0$) is another solution to $Ax = b$, since

$$A(x_0 + v) = Ax_0 + Av = Ax_0 + 0 = b.$$

Therefore we have the following

Is $y \in R(A)$?

No
↓
No solution

YES

Is $N(A) = \{0\}$

YES
↓
UNIQUE
Solution

No
Multiple solutions
exist
(How many ;
Depends on
 F).